Correspondence: Analyzing Figures

Source feature: Figure
Target feature(s):
Automation type: NoDirectMapping -
Relation: Both GMF Figure Descriptors and Sirius container aim to define the visual appear ance in terms of structural composition of diagram elements. However, while GMF supports arbitrary nesting of graphical primitives, Sirius enforces a stricter s eparation between structure (container mappings) and appearance (styles). This m ismatch makes the correspondence inherently approximate and requires manual inte rpretation to preserve the intended visual semantics.

Operators: Ms. Y, Mr. X

Rationale: GMF allows to draw complex figures by defining nested shapes. This could not hap pen in Sirius. This mapping needs supervisor's intuition to have a reasonable ma tch.

Possible approximations (if any): Container Mapping (COMPLEX, MANUAL)

Main Strategy:

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1. [M] supervisors should take the shape (A) of the figure descriptor

2. [M] supervisors have to check if A presents a complex figure by defining nested shap
es (X)

3. [-] if X is true
    3.1. [M] supervisors should split the nested shapes of A into two groups: B1) the shapes
used for only drawing, B2) the shapes used for external access (usually are acce
ssible by a child reference)

    3.2. [-] for every shape (C) in B2
        3.2.1. [M] supervisors should check what canvas feature (D) is referring to/accessing C

        3.2.2. [-] if D is a kind of compartment
            3.2.2.1. [M] supervisors should find the compartment mapping (E) referring to D

            3.2.2.2. [M] supervisors should find E's equivalent container mapping (F)

            3.2.2.3. [M] supervisors should create a style for F in a way that resembels the actual figur
e (Shape Resembalance can be reused)

            3.2.2.4. [M] from A to the most top shape, in a way that matches the figure descriptor in ter
ms of structure, for every shape, supervisors can define a container mapping and
 its style



    3.3. [-] for every shape (G) in B1
        3.3.1. [M] supervisor should find and match G's equivalent mapping (H)

        3.3.2. [M] supervisors should create a style for H in a way that resembels the actual figur
e (Shape Resembalance can be used)
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Structural overview